

Competitive Cybersecurity Learning

Two competitions under one word, and what to do when one of them stops telling people apart

The page states a break, then states what the break costs.

“Six years of population-scale data from a competitive cybersecurity platform document a structural break.”

website, 5-Year Vista, Competitive Cybersecurity Learning, the break

“Access to viable solution pathways has increased faster than the development of underlying expertise. Competitive performance now distinguishes speed of execution rather than depth of problem-solving ability, and the observable gap is between the ability to use AI systems and the ability to understand and appropriately supervise them.”

same page, the mechanism

“If competition no longer measures depth of ability, the instrument the field has used to identify and rank talent has stopped measuring the thing it was trusted to measure.”

same page, why this sits at five years

The page is short because the argument is short. What it needs is a vocabulary, because the phrase at its centre already means something else one project over. In my own files, competitive learning names a machine procedure: a set of strategies that improves by being made to face itself, which is the loop AcceleratePSRO runs and speeds up. On this page it names people learning by competing, and a competition used as a test of what they learned. Both uses are mine, both are current, and the second one is the one that broke.

The overlap is more than a nuisance. The machine sense carries the quantity the human sense has lost. Policy space response oracles have a name for a competition whose entrants have stopped separating, a name for what that costs the analysis, and a name for the repair. Section 3 borrows all three and says where the analogy stops holding.

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1 The census

Where the word appears in my own writing, and in which sense.

competitive cybersecurity training

“what makes a CTF challenge difficult? how can we ensure the effectiveness of competitive cybersecurity training in the era of agent solvers?”

website, Problem-Solving, the branch index

The human sense. The second half of that question and the vista page are the same question at two time horizons, which is why item 76 and the Capture-the-Flag lexicon share entries and have to agree.

competitive tier

“zero; every team in the competitive tier solved every hard challenge”

website, 5-Year Vista, the break, the score-gap row

Also the human sense, and the sentence that does the most work on the page.

strategy population

“assembling a strategy population that represents the full game under a computational budget”

website, AcceleratePSRO, on the strategy exploration problem

The machine sense. No people in it, and the competition is internal to an algorithm.

the era of agent solvers

“The hard part of a CTF is the thinking, not the typing. AI agents can now run the tools, so what is worth teaching and testing is the set of mental moves a solver makes.”

poster draft, SIGCSE Virtual 2026, headline claim

The two senses meet here. The competitors are people, the thing that changed is a machine, and the claim is about what a competition should be scored on once the machine is in the room.

visible problem-solving traces

“a research and dataset-building project that collects public CTFtime writeups and produces structured artifacts for studying visible problem-solving traces”

CTF Solve Trace Atlas, README.md, research purpose

Neither sense. This is the replacement instrument being built, and it is scored on process rather than on outcome by construction, since a writeup that records only the flag is discarded by the pipeline as carrying no evidence.

2 Active

Entries I have to define on the spot, to an educator, to a program officer, or to a reviewer who thinks the vista is a complaint about cheating. All four fields.

2.1 The two competitions

competitive learning

Definition: A scheme in which the improvement of each participant is driven by facing the others, so that what any one of them learns depends on the set they are placed against.

Distinguishes it from: Individual practice against a fixed curriculum, where the difficulty schedule is authored in advance and does not respond to how well anyone is doing. The test is whether removing the other participants changes what gets learned. In competitive learning it does; in a problem set it does not.

Operational test: Ask who the opponent is and whether the opponent moves. A scored ranking against other people, updated as they improve, is the human sense. A set of strategies expanded by best response against the current mixture is the machine sense. A fixed answer key is neither.

In my project: Both. Capture-the-Flag studies people learning under the human sense and asks what happens to it when an agent joins the team. AcceleratePSRO runs the machine sense and asks where the loop can be sped up without losing solution quality. The vista is the place where I have to hold both at once, so every occurrence of the phrase in a paper spanning the two needs a qualifier.

One phrase, and the participant is a person in one project and a policy in the other

On the vista page the participants are teams of students, the payoff is a score on a leaderboard, and the reason to run the competition at all is that the ranking was believed to identify talent. Nothing is being optimised by the organisers. The competition is a measuring device pointed at a population that arrived on its own.

In AcceleratePSRO the participants are policies, the payoff is an entry in an empirical payoff matrix, and the reason to run the loop is to produce the population. Nothing is being measured about anyone who arrived; the arrival is the output.

So the same words name a device for reading a population and a device for manufacturing one. A sentence claiming that competitive learning is effective means, in the first case, that people who compete end up more able, and in the second, that the restricted game converges toward the full game. Neither claim supports the other, and the vista is a case where the first failed while the second was doing fine.

competition as instrument

“the instrument the field has used to identify and rank talent”

website, 5-Year Vista, why this sits at five years

Definition: A competition treated as a measuring device: the score is read as an estimate of a person’s ability, and decisions about admission, hiring and scholarship are taken on that reading.

Distinguishes it from: A competition treated as a training event, where the score is feedback to the participant and nobody outside the room acts on it. The same event can be both, and the failure described on the page hits the instrument role while leaving the training role partly intact.

Operational test: Does anyone who was not in the competition make a decision on the basis of its ranking? If yes, it is an instrument and its validity is a research question. If no, it is practice.

In my project: The vista is a claim about the instrument role specifically. The Capture-the-Flag work is a claim about the training role. Keeping them apart is what stops the vista from reading as an argument that competitions should stop being held.

2.2 The break and what it is a break in

the break

Definition: A discontinuity in a population-scale time series large enough that measurements taken before and after it cannot be pooled, because the quantity being measured is no longer the same quantity.

Distinguishes it from: A trend, which is a change in the value of a stable measurement, and from an outlier year, which reverts. A break is identified by the fact that the measurement procedure has changed meaning, not by the size of the jump.

Operational test: Would you be willing to fit one model across the boundary? If pooling the two sides would produce a number nobody would defend, it is a break.

In my project: Median time to first solve on hard challenges moved from a band of roughly forty to three hundred forty minutes into five minutes; the field-wide solve rate on hard challenges moved from the five to seven percent range to above eighteen percent; and the score gap between first and fiftieth place went to zero. Six years on one platform, so the population and the challenge authoring process are held roughly fixed across the boundary, which is what makes the break attributable to something outside the competition rather than to a change in who entered.

Hard is a design label on one side of the break and a measured property on the other

The Capture-the-Flag lexicon already carries difficulty in three senses. The vista sharpens one of them. When the field-wide solve rate on hard challenges triples, the word hard has not changed: it is still the label the challenge author applied at authoring time, on the basis of what the author expected the field to be able to do. What changed is the field.

A sentence like the hard challenges got easier is therefore false and the wrong shape. The challenges are identical objects. The population's relation to them moved. Any measure of difficulty defined as a solve rate is a measure of a population and a challenge set jointly, and it will report a change in the population as a change in the challenge.

This is the reason the difficulty project asks what makes a challenge difficult rather than which challenges are difficult. The first question has an answer that survives a break; the second does not.

score compression

Definition: The collapse of the spread of scores across a ranked field, to the point where the ranking is decided by a tie-breaker rather than by the scored quantity.

Distinguishes it from: Grade inflation, where scores rise and the spread is preserved, and from a ceiling effect on a single item, where one task saturates while the rest of the instrument still separates. Compression is a property of the whole instrument.

Operational test: Compute the variance of the score across the top of the field. If it is at or near zero and the published ranking is nevertheless ordered, find out what the order is actually made of. On the platform in question the answer is completion time.

In my project: The white paper reports that the top fifty teams in the 2026 competition all achieved identical scores, with time as the differentiator. My page states the same fact as a score gap of zero between first and fiftieth. A ranking produced entirely by a tie-breaker is a ranking on the tie-breaker, and speed of execution is not the construct the ranking was trusted to report.

discriminative power

Definition: The extent to which an instrument assigns different scores to people who differ on the thing it is meant to measure.

Distinguishes it from: Validity, which asks whether the thing being measured is the intended one, and from reliability, which asks whether the same person would score the same twice. An instrument can be perfectly reliable, produce the same score for everyone, and have no discriminative power at all.

Operational test: Take two people you have independent reason to believe differ in ability. Does the instrument put a gap between them? At a solve rate of one hundred percent among the competitive tier, no pair of them can be separated, whatever their difference.

In my project: This is the quantity the vista says was lost, and naming it this way keeps the argument narrow. I am not claiming the competitors are less able than their predecessors. I am claiming the instrument can no longer tell me either way, which is a weaker claim and a worse problem.

2.3 Performance, expertise, and the three ways a skill can be missing

performance

Definition: What a person produces on a task, under whatever conditions and with whatever assistance the task allows.

Distinguishes it from: Expertise, which is a property of the person and is supposed to persist when the conditions change. Performance is a joint property of a person, their tools and the task, and the whole vista is about the widening of the difference.

Operational test: Change the conditions and re-measure. Whatever survives the change was expertise; whatever does not was performance under the old conditions. The reason this test is getting harder to run is in the calibration entry below.

In my project: The title of the white paper the page draws on names the gap between performance and expertise, and my page renames it as the gap between the ability to use AI systems and the ability to understand and supervise them. The second phrasing is the more useful of the two because it says what the missing thing is for, which gives something to teach and something to test.

expertise maintenance

Definition: Keeping a skill available in a workforce over time, against decay in the individuals who hold it and against loss of the conditions under which new people would acquire it.

Distinguishes it from: Training, which is about acquisition, and from retention in the personnel sense, which is about keeping people. Maintenance is about keeping the skill, and it can fail while the roster is full.

Operational test: Ask where the next person who has this skill is going to come from, and what they will practice on. If the answer is a task that is now routinely delegated, the skill is not being maintained regardless of how many people currently hold it.

In my project: The second half of the vista title, and the part that makes it a five-year problem rather than an assessment problem. The competition failure matters because the competition was also the pipeline.

deskilling

Definition: The loss of a skill a person previously held, following a period in which the skill was not exercised because something else performed it.

Distinguishes it from: Two neighbours it gets confused with. Skill formation failure, where the skill was never acquired because the learner delegated during the window in which they would have built it. And ordinary skill decay, where the skill lapses through disuse with no substitute involved. The three have different populations, different remedies, and different evidence.

Operational test: Ask whether the person ever had it. If yes and the lapse follows exposure to a substitute, deskilling. If no and the substitute was present during learning, formation failure. If yes and no substitute is involved, decay.

In my project: The reading table under this heading holds all three and the distinction is not made there, which is a defect I am fixing here rather than in the table. The endoscopy result is deskilling in the strict sense, since the clinicians were competent before exposure. The controlled study on developers learning an unfamiliar asynchronous library is formation failure, since nobody in either arm knew the library at the start. The high-school mathematics result is also formation failure. My page cites the developer study as showing that AI assistance impaired conceptual understanding, code reading and debugging without significant average efficiency gains, and that is a statement about acquisition, not about loss.

The two costs run on different clocks and only one of them shows up in a competition

Deskilling in the strict sense is fast to observe and reversible in principle, because the person can be put back on the task. Formation failure is slow to observe and expensive to reverse, because there is no earlier state to return to and the window in which the skill would have been built has closed.

A competition sees neither directly. It sees performance, which both leave intact or improve. This is why the vista does not argue from the scores at all after the first section: the scores are the symptom that something is unobservable, not the evidence of what happened. The evidence has to come from studies where the assistance is randomised and the outcome is comprehension rather than completion, and there are so far very few of those.

2.4 Knowing how well you are doing

calibration

Definition: Agreement between a person’s estimate of their own performance and their measured performance.

Distinguishes it from: Confidence, which is the estimate alone, and from accuracy, which is the performance alone. Calibration is defined only when both are in hand, which is why it is the first casualty of an instrument that has stopped separating people.

Operational test: Collect the forecast before the task and the estimate after it, then compare both against the measurement. Two gaps, and they can point in different directions.

In my project: The result my page calls the most pointed finding. Sixteen experienced developers across two hundred forty six tasks in repositories they already knew forecast that the tools would cut their completion time by twenty four percent, estimated afterwards that they had been cut by twenty percent, and were in fact slowed by nineteen percent. My page says what that means in one line: they were wrong about their own speed, in the direction that keeps the habit.

control-arm attrition

Definition: Systematic loss of participants or tasks from the arm of a trial that withholds the treatment, produced by the participants’ unwillingness to be in that arm.

Distinguishes it from: Ordinary dropout, which is not selective on the treatment, and from non-compliance, where the participant stays and violates the protocol. Attrition of this kind biases the estimate rather than only shrinking it, and the direction of the bias is knowable.

Operational test: Ask the people who declined why they declined. If the reason names the withheld treatment, the missing tasks are not missing at random and the arm is no longer a control.

In my project: The follow-up to the developer productivity study ran into exactly this and said so: a significant increase in developers choosing not to participate because they did not wish to work without the tools, with thirty to fifty percent of them reporting they had declined to submit particular tasks for that reason, which systematically removes the tasks with the highest expected uplift. The follow-up estimates were withdrawn as unreliable and the design is being changed.

I put this in the lexicon rather than in a footnote because it is the vista’s argument arriving early and in the methods section. The ability to measure what a person can do without assistance is itself a resource, and it is being spent. A five-year program that plans to answer these questions by randomising the tools has to assume that the control arm will be harder to fill every year it waits.

process measurement

“What replaces it has to record process rather than outcome”

website, 5-Year Vista, why this sits at five years

Definition: Scoring a person on the sequence of steps by which they reached an answer, rather than on whether the answer was reached.

Distinguishes it from: Outcome measurement, which reads only the final state, and from time-to-outcome, which reads the final state and a clock and is still an outcome measure. A tie-broken leaderboard is an outcome measure with a clock.

Operational test: Could two participants who both reached the flag receive different scores? If not, the instrument is reading outcomes only, and it will saturate as soon as everyone reaches the flag.

In my project: This is the design requirement the vista hands to the Capture-the-Flag work, and it is why that project is built on six coarse phases and on typed operations between questions rather than on submission logs. It is also why the CTF Solve Trace Atlas keeps a writeup only when the text contains evidence of hypothesis, test, revision and verification, and discards writeups that report only the answer. The last clause of the page adds the second requirement: the replacement has to be run where the conditions can be varied on purpose, which is a classroom and not a leaderboard.

3 What PSRO lends, and where the loan stops

The vista names a competition that has stopped separating its entrants. Empirical game-theoretic analysis has been staring at that object for twenty years, because it is the condition under which its main loop terminates. The vocabulary transfers further than I expected it to, and the places it fails are informative.

best response, and the response oracle

“This challenge of restricted game construction with minimum computational cost (i.e., with the fewest strategies required) is described as the strategy exploration problem”

Bighashdel et al. 2024, Section 2.2

Definition: In PSRO, the subroutine that takes a distribution over the opponents currently in the population and returns a strategy that does as well as possible against it. The loop alternates between solving the restricted game for a mixture and calling the oracle against that mixture.

Distinguishes it from: A learner improving against a fixed opponent. The oracle’s target changes every iteration, and the cost of one call is the dominant cost of the whole procedure, which is what AcceleratePSRO is trying to reduce with imagined rollouts.

Operational test: Is there a step that computes a new strategy against the current set, and does the set grow when it succeeds? Both are needed for the word to apply.

In my project: A CTF competition has this shape read from the other side. The challenge set is the mixture, each team computes a response to it under a time budget, and the response oracle is the team plus its tools. The white paper documents the oracle becoming cheap. In the machine setting a cheap oracle is the entire goal and a published result; in the human setting it is the failure the vista is about, because the population being read is the population of responses.

A cheap best response is the deliverable in one project and the disaster in the other

The AcceleratePSRO page states the aim in one clause: imagined rollouts can stand in for expensive environment interaction when computing a best response inside PSRO. The deliverable

is the crossover point below which imagination stops paying. Cheapening the oracle is the contribution.

On the vista page the oracle got cheap without anyone asking, and every downstream quantity that depended on its cost lost meaning. The two pages therefore assign opposite value to the same event, and they are both right, because in the first case the cost of the oracle was overhead and in the second case it was the signal.

The general form, which is worth stating once because it recurs across my projects: whenever a quantity is used both as a cost to be reduced and as a measurement to be read, some improvement will destroy the measurement. Effort was the measurement here. Nobody wrote that down before it was gone.

a terminated empirical game

Definition: The state PSRO stops in: no new strategy the oracle returns improves on the current mixture, so the restricted game has stopped growing and regret against it is zero.

Distinguishes it from: Convergence to the full game, which is what termination is hoped to indicate but does not guarantee. A run also terminates when the oracle is too weak to find an improvement, and the two are indistinguishable from inside the loop.

Operational test: Ask whether the oracle failed to find an improvement or was unable to look. In PSRO this is the difference between a solution and a budget exhaustion, and it is checked by strengthening the oracle and running again.

In my project: Every team in the competitive tier solving every hard challenge is a terminated game with zero regret. In PSRO that would be reported as a solved restricted game. Read as an instrument, it is the opposite: the analysis has nothing left to say because every entry in the payoff matrix is the same number. The transfer is exact on the arithmetic and inverted on the interpretation, and the reason is that PSRO owns its strategy set while a competition does not. When you can add a strategy you call termination success. When the strategies walk in the door you call it a ceiling.

strategy exploration, read as challenge design

Definition: The problem of choosing which strategies enter the restricted game, given that each admission costs a call to the oracle and that the analysis is only as good as the set.

Distinguishes it from: Solving the restricted game, which is what happens after the set is fixed. Learn Structure supplies the sharp version of the worry: restricting the strategy space is standard rather than cheating, and the restriction error against the full policy space is not bounded.

Operational test: Ask what the analysis would report if a strategy outside the set existed and beat everything in it. If the answer is that the analysis would not notice, you have a strategy exploration problem and not a solver problem.

In my project: Turned around, this is the design brief for a competition that survives cheap oracles. The challenge set is the strategy set, admission is expensive because authoring is expensive, and the instrument fails exactly when the set stops covering what the entrants can do. The unbounded restriction error is the same statement as the vista's: a competition that only contains challenges the field can now solve reports nothing about the field, and there is no bound that says how wrong the report is. Challenge authoring has to outrun the oracle, and the

six-year series says by how much it currently fails to.

exploitability

Definition: How much a player can lose to an opponent who best-responds to them. Over a population, the regret of the mixture against the best response available outside it.

Distinguishes it from: Performance against the strategies already present, which is what a scoreboard reports. A profile can top the scoreboard and be highly exploitable, and that gap is the whole reason the measure exists.

Operational test: Name the best response and compute what it takes. If no best response has been computed, the number on the board is not an exploitability and must not be read as one.

In my project: The word transfers to the instrument rather than to the competitors. A test is exploitable when a strategy exists that scores well on it without possessing the ability it was built to detect, and a field that best responds to the test rather than to the domain will find that strategy, without anyone deciding to cheat. Delegation is that best response. So the vista's claim in one line: the instrument was always exploitable and the exploit was expensive until it was not.

The loan stops at three places, and they are worth stating because the analogy is attractive enough to be over-driven.

- PSRO controls admission to its own population; a competition does not control who enters or what tools they bring. Every conclusion that depends on choosing the population fails to transfer.
- PSRO's payoffs are defined by the game and are stable. A competition's payoffs are authored by people, under time pressure, and the authoring is itself subject to the same assistance the entrants have.
- Regret in PSRO is computed against a full game that exists in principle. In a competition there is no full game to compute against, which is why validity has to be established against an external criterion, such as later job performance, rather than internally.

4 Working

Recognise precisely, look up when needed.

empirical game, restricted game

The payoff matrix estimated by simulating a chosen set of strategies against each other, and the game defined over that set alone. Distinguished from the full game by the restriction, and the whole practice of empirical game-theoretic analysis consists of taking that restriction seriously.

double oracle

The two-player version, in which both sides iteratively learn approximate best responses against each other. Palmer et al. apply it to autonomous cyber defence and record the property that matters here: one call is expensive, which is why they add potential-based reward shaping to expedite it and extend the formulation to multiple response oracles so that several published defenders can be evaluated in one framework. Distinguished from self-play by the fact that both populations are retained.

self-play

A strategy trained against its own current copy. The set has size one at every moment, so there is no diversity to select over. Distinguished from population-based training in the PSRO sense on exactly that point, and the survey draws the contrast before applying the label.

population

Six senses, enumerated in `dict_population_based_training.tex`. This page uses sense six, the sampled population of a defined group, when it says population-scale data, and it means the entrants across six years of one platform. It uses sense two, the strategy population, only inside Section 3. The two share no property except plurality and the file has to keep them apart on every line.

item discrimination

In test theory, the degree to which a single item separates high scorers from low scorers on the instrument as a whole. An item everyone passes has a discrimination index of zero and contributes nothing but time. The hard challenges of the 2026 competition are items whose discrimination went to zero without their content changing.

ceiling effect

Saturation of an instrument at its maximum, so that differences above the ceiling are invisible. Distinguished from score compression, which is the field-level consequence, and from a floor effect at the other end. A ceiling is diagnosed by the shape of the distribution and is fixed by adding harder items, which is why challenge authoring is the operational answer to the vista and not a side task.

construct validity

Whether the instrument measures the construct it names. Distinguished from predictive validity, which asks only whether the score predicts a later outcome and can hold for the wrong reason. The vista's claim is about construct validity: the score still predicts something, and what it predicts is now execution speed.

criterion

The external standard an instrument is validated against. A competition ranking has no internal criterion, which is why the failure described here could persist for a full cycle before it was documented.

stimulated recall

Replaying a participant their own session as a retrieval cue, anchored to timestamps. The method the Capture-the-Flag page settles on, after ruling out concurrent think-aloud as too intrusive for a timed competition, retrospective think-aloud as an invitation to invention, and experience sampling as an interruption.

cognitive task analysis

Structured elicitation of expert reasoning in context, including the critical decision method. Expensive, and dependent on having an expert available, which is the constraint that makes expertise maintenance a methods problem as well as a workforce problem.

agent solver

An autonomous system that attempts a challenge end to end, choosing its own tools and prompts. Distinguished from an assisted human, who remains the one deciding. The distinction is doing real work: in a study of forty one competitors and four autonomous agents at one competition, solve rates for the assisted humans were limited by their own prompting and context specification rather than by the model's reasoning, while the autonomous agents, which set their own prompts, placed second overall.

5 Peripheral

Know roughly where these belong and retrieve when needed.

- **Elo and leaderboard ranking.** Rating systems that assume score differences carry information. They degrade to noise under compression, and the tie-breaker becomes the rating.
- **Deliberate practice.** Structured, effortful repetition with feedback at the edge of ability. The construct the delegation habit removes, and the reason effort deserves to be recorded rather than only minimised.
- **Near and far transfer.** Whether a skill practiced in one setting appears in a distant one. The question underneath any claim that a competition prepares people for operational work.
- **Tacit knowledge.** What experts know and cannot state. Relevant here because it is the part of expertise least likely to be captured by an outcome measure and least likely to be acquired by delegation.
- **Autocurriculum.** A difficulty schedule that emerges from participants improving against each other rather than being authored. The machine sense's answer to the ceiling problem, and unavailable to a competition whose challenge set is written by people once a year.
- **Concept inventory.** A validated instrument targeting conceptual understanding rather than task completion. The nearest existing thing to what the vista says has to be built, and a source of psychometric procedure.

6 Placement on the five dimensions

Using the axes from the dictionary overview page, for competitive learning in its two senses.

- **Object.** Human sense: the behavior of people, and through it their ability. Machine sense: the strategies of a player.
- **Epistemic operation.** Human sense: describe and infer. The competition is run to find out something already true about the entrants. Machine sense: enact. The competition is run to bring the population into existence.
- **Specificity.** Human sense: a particular cohort in a particular year on a particular platform, and the whole argument of the vista is that the year matters. Machine sense: one game, indefinitely.

- **Purpose.** Human sense: defensive evaluation of a workforce pipeline. Machine sense: strategic response.
- **Fidelity.** Human sense: executable, in the sense that the behavior being read is real behavior in a real event. Machine sense: abstract representation.

The pair sits at opposite corners on four of the five axes and shares only the word. That is a stronger separation than any other collision in this lexicon set, and it is the reason the phrase gets qualified every time it appears in writing that touches both projects.

7 What the vocabulary is for

Three consequences follow from the entries above, and they are the reason the vista sits at five years.

The measurement problem comes before the pedagogy problem. Until there is an instrument with discriminative power under assisted conditions, no claim about whether assistance helps or harms a cohort can be settled, including the claims on this page. Process measurement is the requirement, and it means scoring the solve rather than the flag.

The window for building it is closing on its own. Control-arm attrition is already biasing the studies that would answer the question, and the population of people who can be asked to work unassisted shrinks every year. That is a schedule constraint on the research, not only on the workforce.

And the challenge set has to grow faster than the oracle gets cheap. That is the strategy exploration problem stated for people, with the same unbounded restriction error attached, and it says that authoring is where the effort has to go. A competition that only asks what the field can already do reports nothing about the field, and it will keep reporting it in full confidence.

8 Sources

- My own files: the 5-Year Vista page *What Makes it So Difficult? Competitive Cybersecurity Learning and Workforce Expertise Maintenance*; the Capture-the-Flag page and the Problem-Solving branch index; the AcceleratePSRO page; the CTF Solve Trace Atlas `README.md` and `docs/research/framing.md`; the SIGCSE Virtual 2026 poster claim; and the companion entry `dict_population_based_training.tex`, which fixes the six senses of population this file relies on.
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- The deskilling and metacognition rows of the reading table in `content/personal/overview.md`, for the endoscopy result, the high-school mathematics result, procedural skill retention and decay, near and far transfer, deliberate practice, and the psychometric material on concept inventories and on problem-solving times.